

Actuarial Senescence Analysis

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2026-02-27

1. Load and prepare data

```
data <- read.csv(here("data", "processed", "census_data.csv"),
  na = c("", "NA", ".", "?")) %>%
  select(Year, FemID, Site, Age, DeathYear) %>%
  rename(year = Year, femid = FemID, site = Site, death = DeathYear) %>%
  mutate(
    birth = year - Age,
    femid = as.factor(femid),
    site = as.factor(site),
    death = as.integer(death), # NAs introduced here are expected: individuals
    birth = as.integer(birth)) # with no recorded death year are right-censored
                                # (death set to 0 in prepare_basta_input)
data <- data %>% filter(!is.na(femid))
```

2. Helper functions

```
# Prepare BaSTA input matrix from site-level data.
# birth/death NAs are replaced with 0 (BaSTA convention for unknown dates).
# Row order of birthDeath is matched to Y matrix by ID to avoid misalignment.
prepare_basta_input <- function(site_data, study_start, study_end) {
  site_data$birth[is.na(site_data$birth)] <- 0
  site_data$death[is.na(site_data$death)] <- 0

  Y <- CensusToCaptHist(ID = site_data$femid, d = site_data$year, dformat = "yyyy")

  birthDeath <- site_data %>%
    select(femid, birth, death) %>%
    distinct(femid, .keep_all = TRUE)

  # Match row order to Y matrix rather than assuming sorted order
  birthDeath <- birthDeath[match(Y[, 1], birthDeath$femid), ]

  n_year_cols <- ncol(Y) - 1 # minus the ID column
  inputMat <- as.data.frame(cbind(Y, birthDeath[, -1]))
  inputMat <- inputMat[, c(1, n_year_cols + 2, n_year_cols + 3, 2:(n_year_cols + 1))]
```

```

cat("Total individuals:", nrow(inputMat), "\n")
cat("Total capture events:", sum(Y[, -1]), "\n")

DataCheck(inputMat, studyStart = study_start, studyEnd = study_end, silent = FALSE)

return(inputMat)
}

```

COOL HIGHLAND (CENTRAL PLATEAU) ANALYSIS

3. Prepare Plateau data

```

dataCP <- data %>% filter(site == "CP") %>% select(-site)
inputMat_CP <- prepare_basta_input(dataCP, study_start = 2000, study_end = 2024)

```

```

## Total individuals: 1197
## Total capture events: 2539
## No problems were detected with the data.

```

4. Model selection — compare mortality functions (CP)

```

set.seed(123)
multiout_CP <- multibasta(
  object = inputMat_CP, studyStart = 2000, studyEnd = 2024,
  models = c("EX", "GO", "WE", "LO"),
  shapes = c("simple", "Makeham"),
  nsim = 4, niter = 10000, burnin = 2000, thinning = 10, ncpus = 4)

saveRDS(multiout_CP, here("basta_outputs", "multiout_CP.rds"))

```

```

multiout_CP <- readRDS(here("basta_outputs", "multiout_CP.rds"))
summary(multiout_CP, digits = 3)

```

```

##
## Call:
## Covars. structure      : 0
## Minimum age           : fused
## Cat. covars.           :
## Cont. covars.          :
##

```

```
## Model settings:
##      niter      burnin thinning      nsim
##      10000       2001       10         4
##
## DICs:
##      model      shape D.ave D.mode    pD k   DIC DICdiff Rank
## 5      GO Makeham  5302   3821   740 3 6042    0.0    1
## 2      GO  simple  5298   3777   761 2 6059   16.9    2
## 6      WE Makeham  5050   3007 1022 3 6071   29.3    3
## 3      WE  simple  5049   2893 1078 2 6127   84.6    4
## 4      LO  simple  4773    891 1941 3 6714  672.2    5
## 7      LO Makeham  4776    841 1968 4 6744  702.3    6
## 1      EX  simple  6504   4151 1177 1 7681 1639.1    7
```

```
# Result: Gompertz-Makeham selected as best model by DIC
```

5. Validate recapture assumption (CP)

BaSTA assumes constant recapture probability across years by default. We test this by comparing constant vs. time-varying recapture under the best-fitting model (Gompertz-Makeham). Lower DIC supports the better-fitting recapture structure.

```
out_CP_const <- basta(
  object = inputMat_CP, studyStart = 2000, studyEnd = 2024,
  model = "GO", shape = "Makeham", minAge = 2,
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE)

out_CP_tv <- basta(
  object = inputMat_CP, studyStart = 2000, studyEnd = 2024,
  model = "GO", shape = "Makeham", minAge = 2,
  recaptureType = "year",
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE)

saveRDS(out_CP_const, here("basta_outputs", "out_CP_const.rds"))
saveRDS(out_CP_tv, here("basta_outputs", "out_CP_tv.rds"))

out_CP_const <- readRDS(here("basta_outputs", "out_CP_const.rds"))
out_CP_tv <- readRDS(here("basta_outputs", "out_CP_tv.rds"))

cat("CP - Constant recapture DIC:", out_CP_const$DIC["DIC"], "\n")

## CP - Constant recapture DIC: 4842.52

cat("CP - Time-varying recapture DIC:", out_CP_tv$DIC["DIC"], "\n")

## CP - Time-varying recapture DIC: 4821.97
```

```
cat("Constant recapture:\n")
```

```
## Constant recapture:
```

```
print(out_CP_const$coefficients[1:3, ])
```

```
##           Mean      StdErr  Lower95%CI  Upper95%CI  SerAutocorr  UpdateRate
## c    0.009753582 0.009238412  0.0002429403  0.03471263   0.2669797  0.2566944
## b0 -2.110299869 0.101146718 -2.3270426328 -1.92568778   0.6446408  0.2519556
## b1  0.240806202 0.020477414  0.2012331483  0.28177599   0.6357877  0.2508889
## PotScaleReduc
## c          1.000276
## b0          1.000776
## b1          1.000395
```

```
cat("\nTime-varying recapture:\n")
```

```
##
```

```
## Time-varying recapture:
```

```
print(out_CP_tv$coefficients[1:3, ])
```

```
##           Mean      StdErr  Lower95%CI  Upper95%CI  SerAutocorr  UpdateRate
## c    0.009860758 0.00962363  0.000241996  0.03548566   0.2739662  0.2521056
## b0 -2.110105757 0.10211061 -2.323205481 -1.92329622   0.6429345  0.2562556
## b1  0.240505292 0.02048887  0.200251136  0.28114726   0.6310441  0.2506000
## PotScaleReduc
## c          1.000307
## b0          1.003238
## b1          1.002620
```

```
# Time-varying fits best
```

6. Final model — Gompertz-Makeham (CP) with time-varying recapture

```
set.seed(123)
out_CP_GM_final <- basta(
  object = inputMat_CP, studyStart = 2000, studyEnd = 2024,
  model = "GO", shape = "Makeham", minAge = 2,
  recaptureType = "year",
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE, lifeTable = TRUE)

saveRDS(out_CP_GM_final, here("basta_outputs", "out_CP_GM_final.rds"))
```

```
out_CP_GM_final <- readRDS(here("basta_outputs", "out_CP_GM_final.rds"))
summary(out_CP_GM_final, digits = 3)
```

```
##
## Call:
## Model              : GO
## Shape              : Makeham
## Minimum age        : 2
## Covars. structure  : fused
## Cat. covars.       :
## Cont. covars.      :
##
## Model settings:
##      niter    burnin thinning      nsim
##      50000     5001      20         4
##
## Mean Kullback-Leibler
## discrepancy calibration (KLDC):
## KLDC was not calculated due to lack of convergence,
## or because covariates were not included in the model.
##
## Coefficients:
##           Mean   StdErr Lower95%CI Upper95%CI SerAutocorr UpdateRate
## c           0.009818 0.009422  2.63e-04   0.03454    0.2883    0.268
## b0          -2.107567 0.100793 -2.32e+00 -1.92103    0.6384    0.249
## b1           0.240392 0.020397  2.00e-01   0.28021    0.6360    0.253
## lambda      0.000613 0.000615  1.37e-05   0.00225    0.0289    0.241
## pi           0.552027 0.008781  5.35e-01   0.56909    0.1331    1.000
##           PotScaleReduc
## c                   1
## b0                   1
## b1                   1
## lambda              1
## pi                   1
##
## Convergence:
## Appropriate convergence reached for all parameters.
##
## DIC:
## 4833.851
```

```
cat("\nMortality parameters:\n")
```

```
##
## Mortality parameters:
```

```
print(round(out_CP_GM_final$coefficients[1:3, ], 4))
```

```
##           Mean StdErr Lower95%CI Upper95%CI SerAutocorr UpdateRate PotScaleReduc
## c    0.0098 0.0094    0.0003    0.0345    0.2883    0.2682    1.0000
## b0 -2.1076 0.1008   -2.3194   -1.9210    0.6384    0.2486    1.0005
## b1  0.2404 0.0204    0.2002    0.2802    0.6360    0.2535    1.0010
```

```
cat("\nConvergence (Rhat):\n")
```

```
##
## Convergence (Rhat):
```

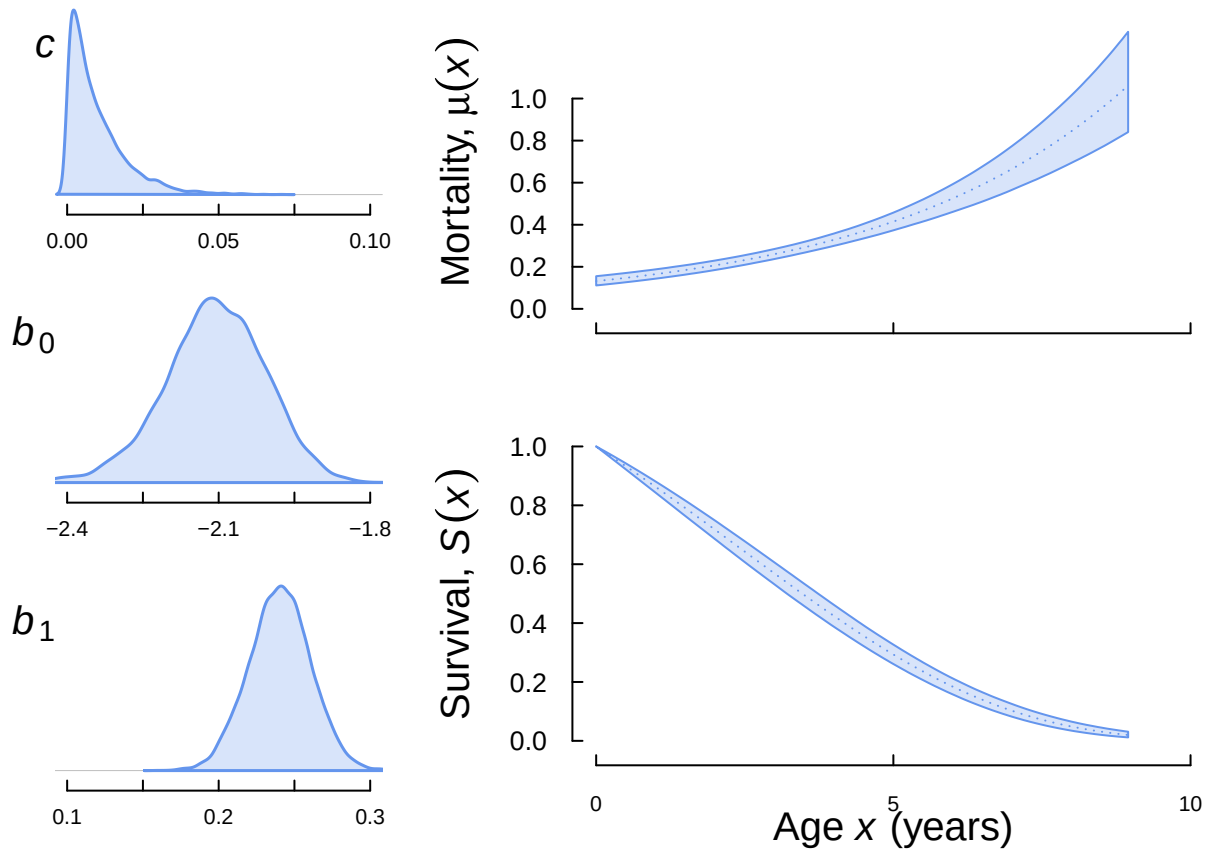
```
print(out_CP_GM_final$convergence)
```

```
##           B           W          Varpl          Rhat
## c      9.440264e-05 8.877274e-05 8.877524e-05 1.000014
## b0     3.335826e-02 1.015143e-02 1.016175e-02 1.000508
## b1     2.289108e-03 4.154009e-04 4.162337e-04 1.001002
## lambda 8.714237e-08 3.777758e-07 3.776466e-07 1.000000
## pi     8.587902e-05 7.711096e-05 7.711486e-05 1.000025
```

```
life_CP <- round(out_CP_GM_final$lifeTable$noCov$Mean, 4)
print(life_CP)
```

```
##    Ages      Nx  Dx    lx    px    qx    Lx    Tx    ex
## 1     0  992.6181   0 1.0000 1.0000 0.0000 1.0000 4.2426 4.2426
## 2     1 1071.4669 234 1.0000 0.7806 0.2194 0.9067 3.2426 3.2426
## 3     2  885.9731 132 0.7806 0.8491 0.1509 0.7189 2.3358 2.9923
## 4     3  797.9249 185 0.6628 0.7735 0.2265 0.5610 1.6170 2.4396
## 5     4  646.0000 158 0.5127 0.7554 0.2446 0.4377 1.0559 2.0597
## 6     5  488.0000 238 0.3873 0.5123 0.4877 0.2916 0.6182 1.5963
## 7     6  250.0000 127 0.1984 0.4921 0.5079 0.1539 0.3266 1.6461
## 8     7  123.0000  48 0.0976 0.6098 0.3902 0.0793 0.1727 1.7685
## 9     8   75.0000  39 0.0595 0.4800 0.5200 0.0439 0.0934 1.5688
## 10    9   36.0000  18 0.0286 0.5000 0.5000 0.0226 0.0496 1.7340
## 11   10   18.0000   6 0.0143 0.6667 0.3333 0.0123 0.0270 1.8872
## 12   11   12.0000   5 0.0095 0.5833 0.4167 0.0084 0.0147 1.5406
## 13   12    7.0000   6 0.0056 0.1429 0.8571 0.0046 0.0062 1.1214
## 14   13    1.0000   1 0.0008 1.0000 0.0000 0.0008 0.0016 2.0000
## 15   14    0.0000   0 0.0008 0.0000 1.0000 0.0008 0.0008 1.0000
```

```
plot(out_CP_GM_final, col = "cornflowerblue", plot.type = "fancy")
```



WARM LOWLAND (ORFORD) ANALYSIS

7. Prepare Orford data

```
dataOR <- data %>% filter(site == "OR") %>% select(-site)
inputMat_OR <- prepare_basta_input(dataOR, study_start = 2000, study_end = 2024)
```

```
## Total individuals: 1005
## Total capture events: 2298
## No problems were detected with the data.
```

8. Model selection — compare mortality functions (OR)

Fit all combinations of mortality model x shape to identify the best-fitting parametric survival model. This is a coarse screen using shorter MCMC chains.

```

set.seed(123)
multiout_OR <- multibasta(
  object = inputMat_OR, studyStart = 2000, studyEnd = 2024,
  models = c("EX", "GO", "WE", "LO"),
  shapes = c("simple", "Makeham"),
  nsim = 4, niter = 10000, burnin = 2000, thinning = 10, ncpus = 4)

saveRDS(multiout_OR, here("basta_outputs", "multiout_OR.rds"))

```

```

multiout_OR <- readRDS(here("basta_outputs", "multiout_OR.rds"))
summary(multiout_OR, digits = 3)

```

```

##
## Call:
## Covars. structure      : 0
## Minimum age           : fused
## Cat. covars.          :
## Cont. covars.         :
##
## Model settings:
##      niter    burnin thinning    nsim
##      10000     2001      10        4
##
## DICs:
##   model  shape D.ave D.mode   pD k   DIC DICdiff Rank
## 3    WE  simple 4377  3429  474 2 4852    0.0    1
## 6    WE Makeham 4379  3410  484 3 4863   11.2    2
## 5    GO Makeham 4584  3947  318 3 4902   50.5    3
## 2    GO  simple 4581  3926  328 2 4909   57.3    4
## 4    LO  simple 4178  1956 1111 3 5289  436.9    5
## 7    LO Makeham 4181  1960 1110 4 5291  439.7    6
## 1    EX  simple 5077  3985  546 1 5623  770.9    7

```

Result: WE simple selected as best model by DIC (4852), then WE Makeham (4863)

9. Validate recapture assumption (OR)

BaSTA assumes constant recapture probability across years. Before running the final model, we test this by comparing the DIC of the best mortality model (Weibull simple) under constant vs. time-varying recapture. Lower DIC supports the better-fitting recapture structure.

```

out_OR_WE_const <- basta(
  object = inputMat_OR, studyStart = 2000, studyEnd = 2024,
  model = "WE", shape = "simple", minAge = 2,
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE)

out_OR_WE_tv <- basta(
  object = inputMat_OR, studyStart = 2000, studyEnd = 2024,
  model = "WE", shape = "simple", minAge = 2,

```



```

recaptureType = "year",
nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
updateJumps = TRUE)

saveRDS(out_OR_WE_const, here("basta_outputs", "out_OR_WE_const.rds"))
saveRDS(out_OR_WE_tv, here("basta_outputs", "out_OR_WE_tv.rds"))

out_OR_WE_const <- readRDS(here("basta_outputs", "out_OR_WE_const.rds"))
out_OR_WE_tv <- readRDS(here("basta_outputs", "out_OR_WE_tv.rds"))

cat("OR - Constant recapture DIC:", out_OR_WE_const$DIC["DIC"], "\n")

```

```
## OR - Constant recapture DIC: 3453.584
```

```
cat("OR - Time-varying recapture DIC:", out_OR_WE_tv$DIC["DIC"], "\n")
```

```
## OR - Time-varying recapture DIC: 3396.389
```

```
cat("Constant recapture:\n")
```

```
## Constant recapture:
```

```
print(out_OR_WE_const$coefficients[1:3, ])
```

```
##           Mean      StdErr  Lower95%CI  Upper95%CI  SerAutocorr
## b0      1.5561329710 0.0668754541 1.427675e+00 1.689752711 0.2910156
## b1      0.2491229471 0.0082653609 2.335458e-01 0.265815655 0.2287633
## lambda 0.0008393313 0.0008466091 2.072042e-05 0.003107147 0.0121618
##           UpdateRate PotScaleReduc
## b0      0.2495500      1.000287
## b1      0.2420778      1.000318
## lambda 0.2612333      1.000000
```

```
cat("\nTime-varying recapture:\n")
```

```
##
```

```
## Time-varying recapture:
```

```
print(out_OR_WE_tv$coefficients[1:3, ])
```

```
##           Mean      StdErr  Lower95%CI  Upper95%CI  SerAutocorr
## b0      1.5546839889 0.0656947239 1.428742e+00 1.688974001 0.28166572
## b1      0.2490823035 0.0082367135 2.333290e-01 0.265727724 0.22386227
## lambda 0.0008515662 0.0008579285 2.137801e-05 0.003118169 0.02287089
##           UpdateRate PotScaleReduc
## b0      0.2474944      1.000835
## b1      0.2384222      1.000343
## lambda 0.2578222      1.000000
```

```
# Time-varying fits best
```

10. Final model — Weibull simple (OR) with time-varying recapture

Run with longer chains for robust posterior estimates and life table derivation.

```
set.seed(123)
out_OR_WE_final <- basta(
  object = inputMat_OR, studyStart = 2000, studyEnd = 2024,
  model = "WE", shape = "simple", minAge = 2,
  recaptureType = "year",
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE, lifeTable = TRUE)

saveRDS(out_OR_WE_final, here("basta_outputs", "out_OR_WE_final.rds"))

out_OR_WE_final <- readRDS(here("basta_outputs", "out_OR_WE_final.rds"))
summary(out_OR_WE_final, digits = 3)
```

```
##
## Call:
## Model                : WE
## Shape                 : simple
## Minimum age           : 2
## Covars. structure     : fused
## Cat. covars.          :
## Cont. covars.         :
##
## Model settings:
##   niter  burnin thinning  nsim
##  50000   5001     20      4
##
## Mean Kullback-Leibler
## discrepancy calibration (KLDC):
## KLDC was not calculated due to lack of convergence,
## or because covariates were not included in the model.
##
## Coefficients:
##           Mean   StdErr Lower95%CI Upper95%CI SerAutocorr UpdateRate
## b0      1.556389 0.066949   1.428562   1.69257    0.2771    0.258
## b1      0.248885 0.008137   0.233540   0.26536    0.2051    0.248
## lambda  0.000832 0.000829   0.000021   0.00312    0.0335    0.242
## pi      0.632544 0.009728   0.613414   0.65138    0.1085    1.000
##           PotScaleReduc
## b0                1
## b1                1
## lambda            1
## pi                1
##
```

```
## Convergence:
## Appropriate convergence reached for all parameters.
##
## DIC:
## 3426.503
```

```
cat("\nMortality parameters:\n")
```

```
##
## Mortality parameters:
```

```
print(round(out_OR_WE_final$coefficients[1:3, ], 4))
```

```
##           Mean StdErr Lower95%CI Upper95%CI SerAutocorr UpdateRate PotScaleReduc
## b0      1.5564 0.0669    1.4286    1.6926    0.2771    0.2580    1.0000
## b1      0.2489 0.0081    0.2335    0.2654    0.2051    0.2476    1.0003
## lambda 0.0008 0.0008    0.0000    0.0031    0.0335    0.2423    1.0000
```

```
cat("\nConvergence (Rhat):\n")
```

```
##
## Convergence (Rhat):
```

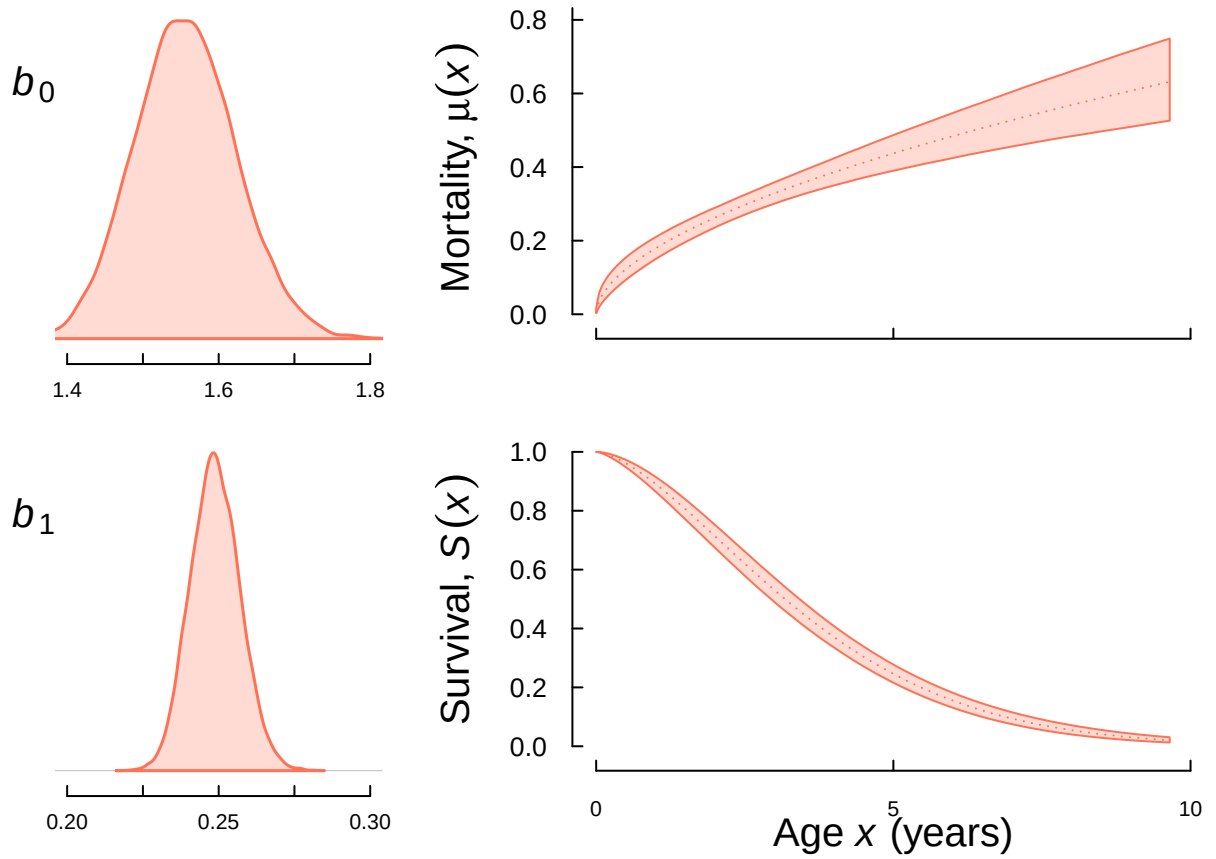
```
print(out_OR_WE_final$convergence)
```

```
##           B           W          Varpl      Rhat
## b0      1.928475e-03 4.483073e-03 4.481937e-03 1.000000
## b1      1.630847e-04 6.618370e-05 6.622677e-05 1.000325
## lambda 3.754796e-07 6.868170e-07 6.866786e-07 1.000000
## pi      1.930555e-04 9.459317e-05 9.463693e-05 1.000231
```

```
life_OR <- round(out_OR_WE_final$lifeTable$noCov$Mean, 4)
print(life_OR)
```

```
##    Ages      Nx  Dx      lx      px      qx      Lx      Tx      ex
## 1     0 799.9448   0 1.0000 1.0000 0.0000 1.0000 3.9477 3.9477
## 2     1 870.2027 280 1.0000 0.6767 0.3233 0.9091 2.9477 2.9477
## 3     2 662.0479 111 0.6767 0.8390 0.1610 0.6099 2.0386 3.0126
## 4     3 598.3112 163 0.5678 0.7270 0.2730 0.4880 1.4287 2.5164
## 5     4 451.0000 146 0.4127 0.6778 0.3222 0.3395 0.9407 2.2792
## 6     5 305.0000 118 0.2798 0.6132 0.3868 0.2322 0.6013 2.1492
## 7     6 187.0000  66 0.1715 0.6471 0.3529 0.1449 0.3690 2.1511
## 8     7 121.0000  54 0.1110 0.5538 0.4462 0.0904 0.2241 2.0191
## 9     8  67.0000  23 0.0615 0.6569 0.3431 0.0512 0.1337 2.1752
## 10    9  44.0000  13 0.0404 0.7045 0.2955 0.0358 0.0826 2.0444
## 11   10  31.0000  15 0.0285 0.5161 0.4839 0.0229 0.0467 1.6425
## 12   11  16.0000   8 0.0147 0.5000 0.5000 0.0118 0.0238 1.6214
## 13   12   8.0000   2 0.0073 0.7500 0.2500 0.0066 0.0120 1.6385
## 14   13   6.0000   5 0.0055 0.1667 0.8333 0.0036 0.0055 0.9935
## 15   14   1.0000   1 0.0009 1.0000 0.0000 0.0009 0.0018 2.0000
## 16   15   0.0000   0 0.0009 0.0000 1.0000 0.0009 0.0009 1.0000
```

```
plot(out_OR_WE_final, col = "coral1", plot.type = "fancy")
```



CROSS-POPULATION COMPARISON

11. Mortality hazard comparison figure

```
extract_mort_curve_full <- function(basta_out, site_name, max_age = 12) {
  ages <- basta_out$x
  idx <- ages <= max_age
  data.frame(
    age = ages[idx],
    mean = basta_out$mort$nocov["Mean", idx],
    lower = basta_out$mort$nocov["2.5%", idx],
    upper = basta_out$mort$nocov["97.5%", idx],
    site = site_name)
}
```

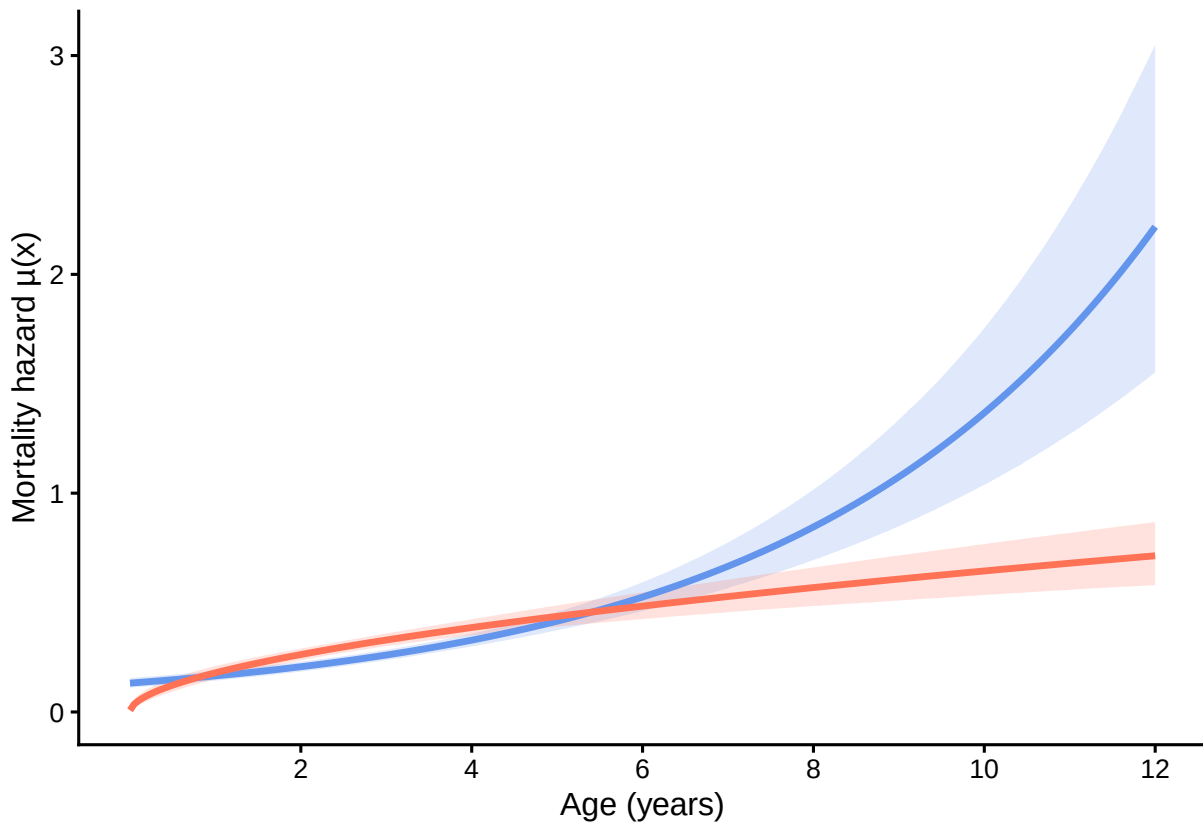
```

haz_data <- bind_rows(
  extract_mort_curve_full(out_OR_WE_final, "Warm lowland"),
  extract_mort_curve_full(out_CP_GM_final, "Cool highland"))

p_haz_overlay <- ggplot(haz_data, aes(x = age, y = mean, color = site, fill = site)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = site_colors) +
  scale_fill_manual(values = site_colors) +
  scale_x_continuous(breaks = seq(2, 12, 2)) +
  labs(y = "Mortality hazard  $\mu(x)$ ", x = "Age (years)") +
  theme_publication

p_haz_overlay

```



```

ggsave(here("outputs", "figures", "Figure_mortality_comparison.png"),
  p_haz_overlay, width = 6, height = 5, dpi = 600)
ggsave(here("outputs", "figures", "Figure_mortality_comparison.pdf"),
  p_haz_overlay, width = 6, height = 5)

```

12. Life expectancy and mortality at key ages

```
# 1. Life expectancy at maturity from life tables
life_OR_WE <- out_OR_WE_final$lifeTable$noCov$Mean
life_CP_GM <- out_CP_GM_final$lifeTable$noCov$Mean

le_OR <- life_OR_WE$ex[life_OR_WE$Ages == 2]
le_CP <- life_CP_GM$ex[life_CP_GM$Ages == 2]
cat("Life expectancy at age 2 - Warm lowland:", round(le_OR, 2), "\n")
```

```
## Life expectancy at age 2 - Warm lowland: 3.01
```

```
cat("Life expectancy at age 2 - Cool highland:", round(le_CP, 2), "\n")
```

```
## Life expectancy at age 2 - Cool highland: 2.99
```

```
# 2. Mortality at specific ages from posterior
ages_of_interest <- c(2, 4, 6, 8, 10, 12)

extract_mort_at_ages <- function(basta_out, ages_of_interest) {
  all_ages <- basta_out$x
  idx <- match(ages_of_interest, all_ages)
  data.frame(
    age = ages_of_interest,
    mean = basta_out$mort$nocov["Mean", idx],
    lower = basta_out$mort$nocov["2.5%", idx],
    upper = basta_out$mort$nocov["97.5%", idx])
}

mort_OR_ages <- extract_mort_at_ages(out_OR_WE_final, ages_of_interest)
mort_CP_ages <- extract_mort_at_ages(out_CP_GM_final, ages_of_interest)

cat("\nWarm lowland mortality at key ages (Weibull simple):\n")
```

```
##
## Warm lowland mortality at key ages (Weibull simple):
```

```
print(round(mort_OR_ages, 4))
```

```
##   age   mean  lower  upper
## 1    2 0.2626 0.2358 0.2896
## 2    4 0.3860 0.3500 0.4237
## 3    6 0.4840 0.4246 0.5469
## 4    8 0.5686 0.4841 0.6607
## 5   10 0.6444 0.5347 0.7674
## 6   12 0.7140 0.5799 0.8675
```

```
cat("\nCool highland mortality at key ages (Gompertz-Makeham):\n")
```

```
##
## Cool highland mortality at key ages (Gompertz-Makeham):
```

```
print(round(mort_CP_ages, 4))
```

```
##   age   mean  lower  upper
## 1    2 0.2069 0.1848 0.2297
## 2    4 0.3283 0.2996 0.3581
## 3    6 0.5253 0.4611 0.5930
## 4    8 0.8456 0.6947 1.0163
## 5   10 1.3671 1.0384 1.7545
## 6   12 2.2178 1.5524 3.0503
```

```
# 3. Fold-increase in mortality from age 2 to age 12
```

```
mort_OR_age2 <- mort_OR_ages$mean[mort_OR_ages$age == 2]
mort_OR_age12 <- mort_OR_ages$mean[mort_OR_ages$age == 12]
mort_CP_age2 <- mort_CP_ages$mean[mort_CP_ages$age == 2]
mort_CP_age12 <- mort_CP_ages$mean[mort_CP_ages$age == 12]
```

```
cat("\nFold-increase in mortality age 2 to 12:\n")
```

```
##
## Fold-increase in mortality age 2 to 12:
```

```
cat("Warm lowland:", round(mort_OR_age12 / mort_OR_age2, 2), "\n")
```

```
## Warm lowland: 2.72
```

```
cat("Cool highland:", round(mort_CP_age12 / mort_CP_age2, 2), "\n")
```

```
## Cool highland: 10.72
```

```
# 4. Confirm senescence in OR: Weibull b1 > 1 indicates increasing hazard
```

```
cat("\nWeibull simple parameters (OR):\n")
```

```
##
## Weibull simple parameters (OR):
```

```
print(round(out_OR_WE_final$coefficients[1:2,
      c("Mean", "Lower95%CI", "Upper95%CI")], 4))
```

```
##      Mean Lower95%CI Upper95%CI
## b0 1.5564      1.4286      1.6926
## b1 0.2489      0.2335      0.2654
```

```
# b1 > 1 = senescence; check CI excludes 1
```

```
# 5. Posterior overlap on mortality trajectories at key ages
```

```
overlap_check <- data.frame(
```

```

age      = ages_of_interest,
OR_mean  = mort_OR_ages$mean,
OR_lower = mort_OR_ages$lower,
OR_upper = mort_OR_ages$upper,
CP_mean  = mort_CP_ages$mean,
CP_lower = mort_CP_ages$lower,
CP_upper = mort_CP_ages$upper)

overlap_check$CIs_overlap <-
  overlap_check$OR_lower < overlap_check$CP_upper &
  overlap_check$CP_lower < overlap_check$OR_upper

print(round(overlap_check, 4))

```

```

##   age OR_mean OR_lower OR_upper CP_mean CP_lower CP_upper CIs_overlap
## 1  2  0.2626  0.2358  0.2896  0.2069  0.1848  0.2297           0
## 2  4  0.3860  0.3500  0.4237  0.3283  0.2996  0.3581           1
## 3  6  0.4840  0.4246  0.5469  0.5253  0.4611  0.5930           1
## 4  8  0.5686  0.4841  0.6607  0.8456  0.6947  1.0163           0
## 5 10  0.6444  0.5347  0.7674  1.3671  1.0384  1.7545           0
## 6 12  0.7140  0.5799  0.8675  2.2178  1.5524  3.0503           0

```

13. Median lifespan (age at which survival drops below 0.5)

```

derive_median_lifespan <- function(basta_out, site_name) {
  ages      <- basta_out$x
  mort_mean <- basta_out$mort$nocov["Mean", ]

  # Cumulative survival from hazard (numerical integration)
  surv <- exp(-cumsum(mort_mean * c(ages[1], diff(ages))))

  # Age at which survival first drops below 0.5
  if (any(surv <= 0.5)) {
    median_ls <- ages[which(surv <= 0.5)[1]]
  } else {
    median_ls <- max(ages)
    warning(site_name, ": survival never drops below 0.5 within observed age range")
  }

  cat(site_name, "median lifespan:", round(median_ls, 2), "years\n")
  return(median_ls)
}

median_OR <- derive_median_lifespan(out_OR_WE_final, "Warm lowland")

```

```
## Warm lowland median lifespan: 3.15 years
```

```
median_CP <- derive_median_lifespan(out_CP_GM_final, "Cool highland")
```

```
## Cool highland median lifespan: 3.5 years
```

TABLES FOR MANUSCRIPT

14. Parameter tables — presented separately as models differ between populations

```
# OR: Weibull simple (b0, b1)
# CP: Gompertz-Makeham (b0, b1, c)
# These are not directly comparable so are presented as separate tables

params_OR <- as.data.frame(out_OR_WE_final$coefficients) %>%
  rownames_to_column("Parameter") %>%
  filter(Parameter %in% c("b0", "b1")) %>%
  select(Parameter, Mean, StdErr, Lower = "Lower95%CI", Upper = "Upper95%CI",
         Rhat = "PotScaleReduc") %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

params_CP <- as.data.frame(out_CP_GM_final$coefficients) %>%
  rownames_to_column("Parameter") %>%
  filter(Parameter %in% c("b0", "b1", "c")) %>%
  select(Parameter, Mean, StdErr, Lower = "Lower95%CI", Upper = "Upper95%CI",
         Rhat = "PotScaleReduc") %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

ft_OR <- flextable(params_OR) %>%
  set_header_labels(StdErr = "SE", Lower = "Lower 95% CI",
                   Upper = "Upper 95% CI", Rhat = "R-hat") %>%
  set_caption("Warm lowland (Weibull simple) mortality parameters.") %>%
  theme_vanilla() %>%
  align(align = "center", part = "all") %>%
  align(j = 1, align = "left", part = "body") %>%
  autofit()

ft_CP <- flextable(params_CP) %>%
  set_header_labels(StdErr = "SE", Lower = "Lower 95% CI",
                   Upper = "Upper 95% CI", Rhat = "R-hat") %>%
  set_caption("Cool highland (Gompertz-Makeham) mortality parameters.") %>%
  theme_vanilla() %>%
  align(align = "center", part = "all") %>%
  align(j = 1, align = "left", part = "body") %>%
  autofit()

ft_OR
```

Table 1: Warm lowland (Weibull simple) mortality parameters.

Parameter	Mean	SE	Lower 95% CI	Upper 95% CI	R-hat
b0	1.556	0.067	1.429	1.693	1
b1	0.249	0.008	0.234	0.265	1

ft_CP

Table 2: Cool highland (Gompertz-Makeham) mortality parameters.

Parameter	Mean	SE	Lower 95% CI	Upper 95% CI	R-hat
c	0.010	0.009	0.000	0.035	1.000
b0	-2.108	0.101	-2.319	-1.921	1.001
b1	0.240	0.020	0.200	0.280	1.001

```
save_as_docx(ft_OR, path = here("outputs", "tables", "Table_params_warm_lowland.docx"))
save_as_docx(ft_CP, path = here("outputs", "tables", "Table_params_cool_highland.docx"))
```

15. Life tables (Supplementary Tables S12–S13)

```
for (info in list(
  list(tbl = life_OR, pop = "Warm lowland", cap = "Table S12",
    file = here("outputs", "tables", "Table_S12_lifetable_warm_lowland.docx")),
  list(tbl = life_CP, pop = "Cool highland", cap = "Table S13",
    file = here("outputs", "tables", "Table_S13_lifetable_cool_highland.docx")))) {

  tbl <- as.data.frame(info$tbl)
  tbl$Population <- info$pop

  ft_lt <- flextable(tbl) %>%
    set_caption(paste0(info$cap, ". Life table for ", info$pop,
      " population.")) %>%
    theme_vanilla() %>%
    align(align = "center", part = "all") %>%
    autofit()

  save_as_docx(ft_lt, path = info$file)
}
```

SUPPLEMENTARY MATERIAL

16. MCMC convergence diagnostics (Figure S1)

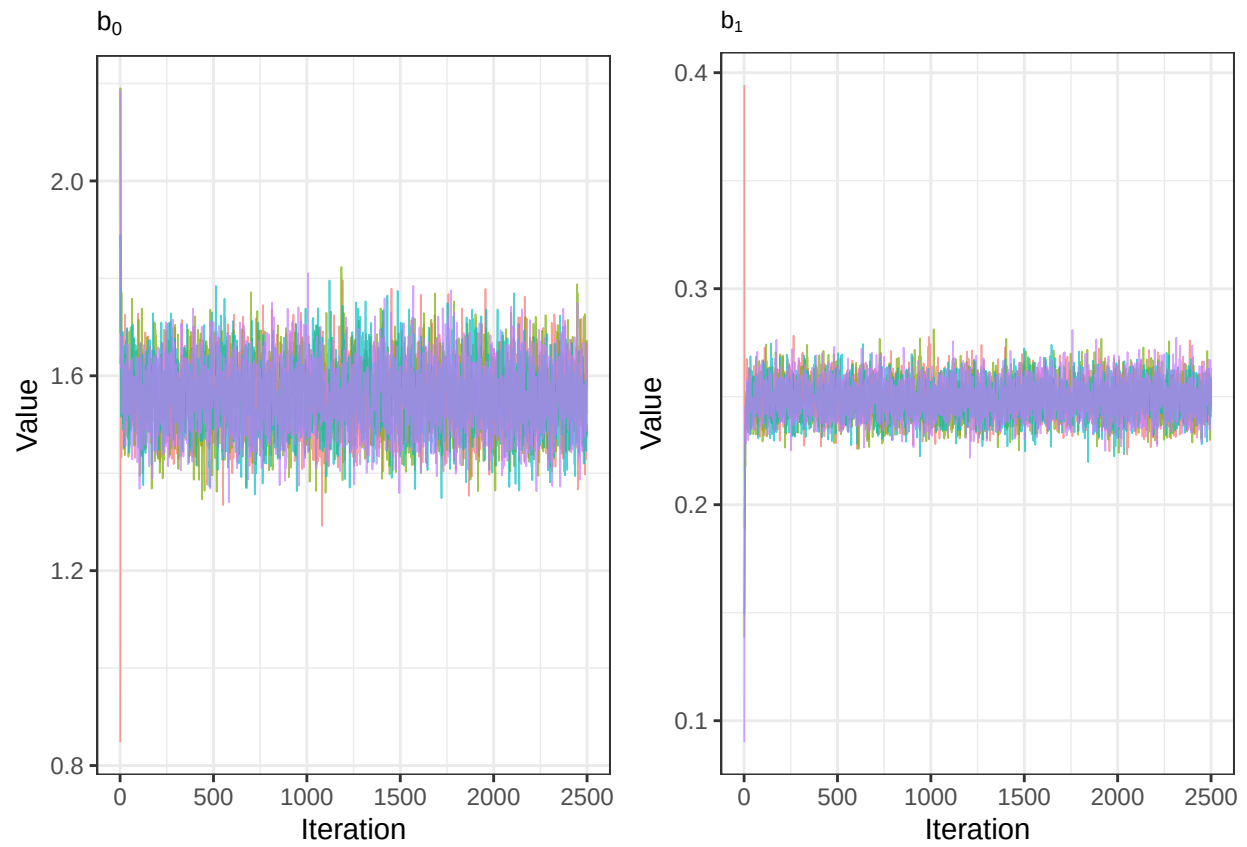
```
extract_trace <- function(model, param) {
  traces <- model$runs
  do.call(rbind, lapply(1:4, function(i) {
    theta_cols <- colnames(traces[[i]]$theta)
    if (param %in% theta_cols) {
      values <- traces[[i]]$theta[, param]
    } else if (param %in% names(traces[[i]])) {
      values <- traces[[i]][[param]][, 1]
    } else {
      stop(paste("Parameter", param, "not found"))
    }
    data.frame(iteration = seq_along(values), value = values,
               chain = paste0("Chain ", i))
  )))
}

make_trace_plot <- function(model, param, title = NULL, subtitle = NULL) {
  ggplot(extract_trace(model, param),
    aes(x = iteration, y = value, color = chain)) +
  geom_line(alpha = 0.7, linewidth = 0.3) +
  labs(title = title, subtitle = subtitle, x = "Iteration", y = "Value") +
  theme_bw() +
  theme(legend.position = "none",
        plot.title = element_text(size = 10),
        plot.subtitle = element_text(size = 9))
}

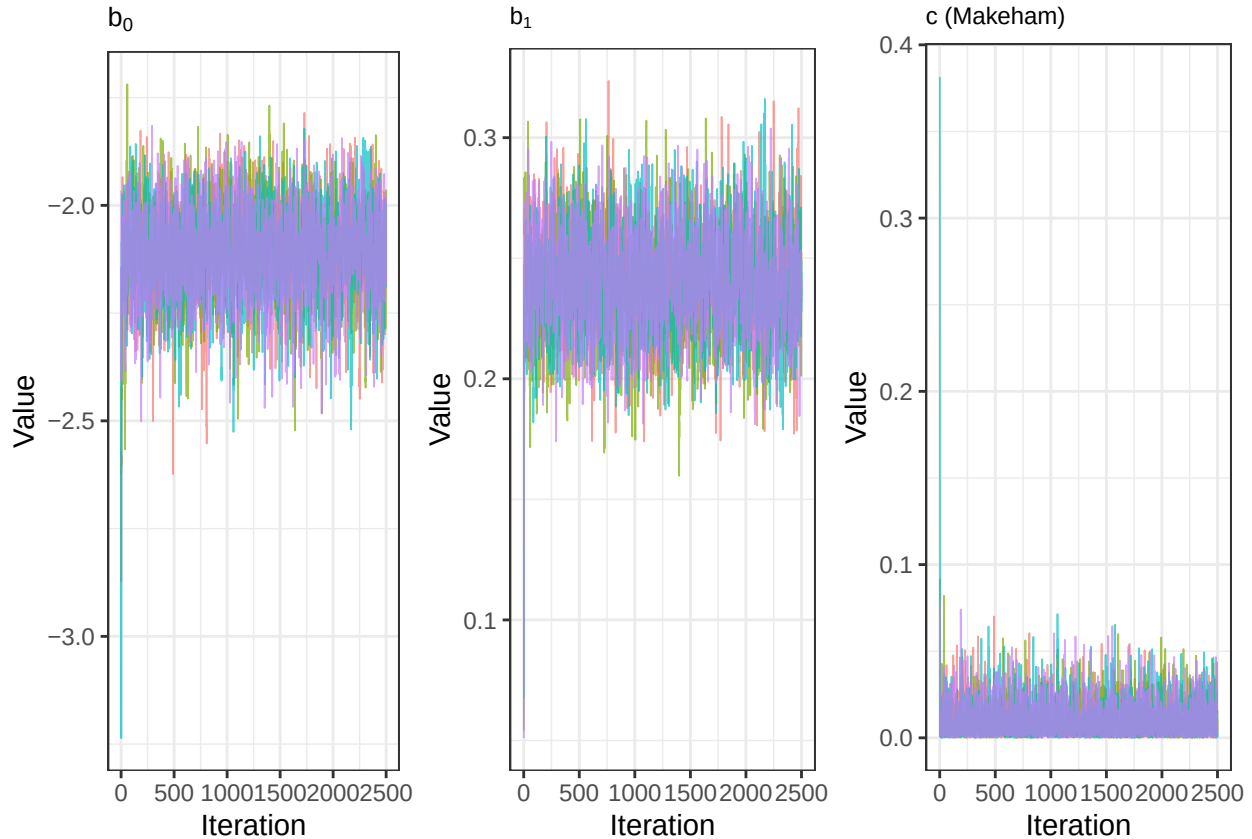
# OR: Weibull simple - trace plots for b0, b1
p1 <- make_trace_plot(out_OR_WE_final, "b0", expression(b[0]))
p2 <- make_trace_plot(out_OR_WE_final, "b1", subtitle = expression(b[1]))

# CP: Gompertz-Makeham - trace plots for b0, b1, c
p3 <- make_trace_plot(out_CP_GM_final, "b0", expression(b[0]))
p4 <- make_trace_plot(out_CP_GM_final, "b1", subtitle = expression(b[1]))
p5 <- make_trace_plot(out_CP_GM_final, "c", subtitle = "c (Makeham)")

fig_S1 <- grid.arrange(p1, p2, ncol = 2, nrow = 1)
```



```
fig_S2 <- grid.arrange( p3, p4, p5, ncol = 3, nrow = 1)
```



```
ggsave(here("outputs", "figures", "Figure_S1_convergence_OR.png"),
       fig_S1, width = 12, height = 7, dpi = 300)
ggsave(here("outputs", "figures", "Figure_S1_convergence_OR.pdf"),
       fig_S1, width = 12, height = 7)
ggsave(here("outputs", "figures", "Figure_S2_convergence_CP.png"),
       fig_S2, width = 12, height = 7, dpi = 300)
ggsave(here("outputs", "figures", "Figure_S2_convergence_CP.pdf"),
       fig_S2, width = 12, height = 7)
```

17. OR model robustness: Weibull simple vs Gompertz-Makeham

Weibull simple was the best fit for OR (Section 8). We tested Gompertz-Makeham to assess whether it could be used instead to allow direct parameter comparison with CP. However, the two models produce divergent mortality trajectories over biologically relevant ages (≤ 14 years), confirming that Gompertz-Makeham is a poor fit for OR and that population-specific best-fitting models must be used. Cross-population comparisons therefore rely on model-agnostic outputs (life expectancy, mortality at key ages) rather than parameter estimates.

```
out_OR_GM <- basta(
  object = inputMat_OR, studyStart = 2000, studyEnd = 2024,
  model = "GO", shape = "Makeham", minAge = 2,
  recaptureType = "year",
  nsim = 4, niter = 50000, burnin = 5001, thinning = 20,
  updateJumps = TRUE, lifeTable = TRUE)
```

```

saveRDS(out_OR_GM, here("basta_outputs", "out_OR_GM.rds"))

out_OR_GM <- readRDS(here("basta_outputs", "out_OR_GM.rds"))
out_OR_WE <- readRDS(here("basta_outputs", "out_OR_WE_final.rds"))

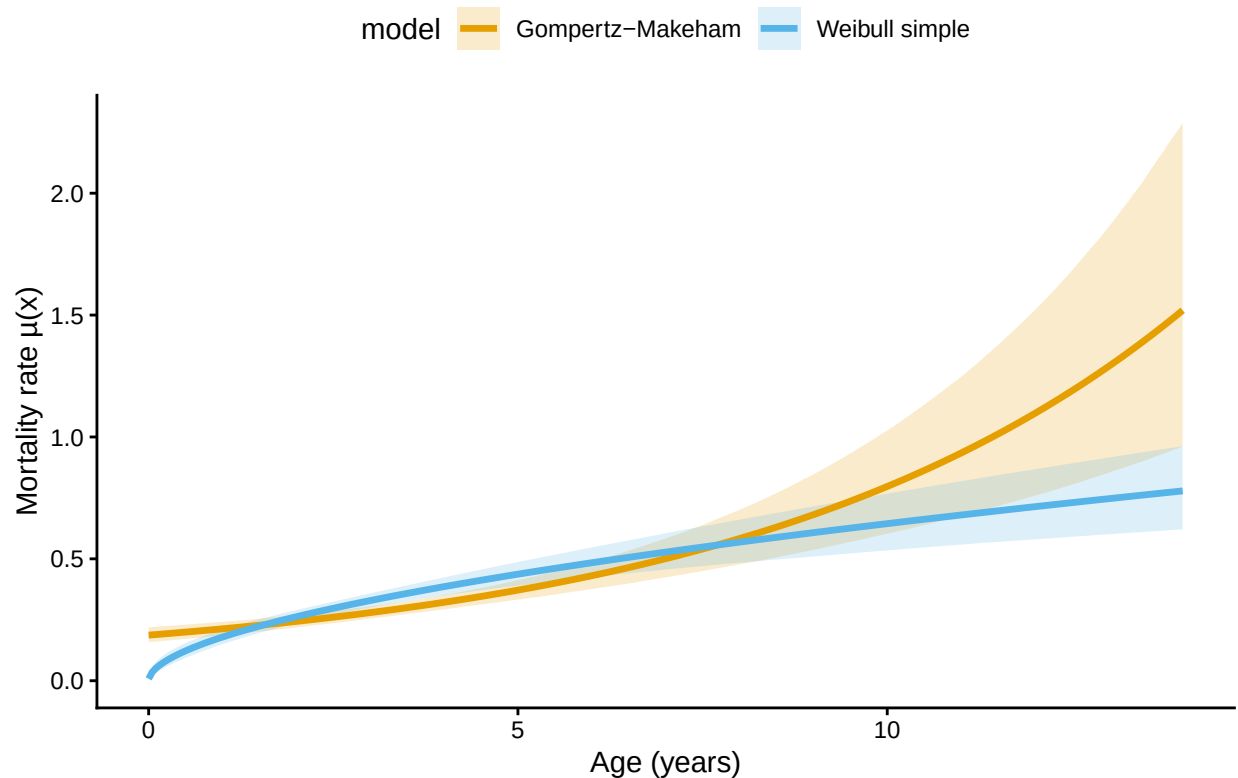
extract_mort_curve <- function(basta_out, model_label, max_age = 14) {
  data.frame(
    age      = basta_out$x,
    mortality = basta_out$mort$nocov["Mean", ],
    lower     = basta_out$mort$nocov["2.5%", ],
    upper     = basta_out$mort$nocov["97.5%", ],
    model     = model_label) %>%
  filter(age <= max_age)
}

mort_compare <- bind_rows(
  extract_mort_curve(out_OR_GM, "Gompertz-Makeham"),
  extract_mort_curve(out_OR_WE, "Weibull simple"))

ggplot(mort_compare, aes(x = age, y = mortality, color = model, fill = model)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = c("Gompertz-Makeham" = "#E69F00", "Weibull simple" = "#56B4E9")) +
  scale_fill_manual(values = c("Gompertz-Makeham" = "#E69F00", "Weibull simple" = "#56B4E9")) +
  labs(x = "Age (years)", y = "Mortality rate \u03bc(x)",
       title = "Warm lowland: Gompertz-Makeham vs Weibull simple (ages 0-14)") +
  theme_classic() +
  theme(legend.position = "top")

```

Warm lowland: Gompertz–Makeham vs Weibull simple (ages 0–14)



```
# Quantify divergence between models
mort_wide <- inner_join(
  extract_mort_curve(out_OR_GM, "GM") %>% select(age, mort_GM = mortality),
  extract_mort_curve(out_OR_WE, "WE") %>% select(age, mort_WE = mortality),
  by = "age")

cat("Correlation (ages 0-14):", round(cor(mort_wide$mort_GM, mort_wide$mort_WE), 4), "\n")
```

```
## Correlation (ages 0-14): 0.9024
```

```
cat("Mean abs % difference:",
  round(mean(abs(mort_wide$mort_GM - mort_wide$mort_WE) /
    ((mort_wide$mort_GM + mort_wide$mort_WE) / 2)) * 100, 2), "%\n")
```

```
## Mean abs % difference: 23.99 %
```

```
# Both models detect senescence (b1 > 0 for GM; b1 > 1 for Weibull)
cat("\nGompertz-Makeham b1 (OR):\n")
```

```
##
```

```
## Gompertz-Makeham b1 (OR):
```

```
print(out_OR_GM$coefficients["b1", c("Mean", "Lower95%CI", "Upper95%CI")])
```

```
##           Mean Lower95%CI Upper95%CI
## 0.1638444 0.1185512 0.2230617
```

```
cat("Weibull simple b1 (OR):\n")
```

```
## Weibull simple b1 (OR):
```

```
print(out_OR_WE$coefficients["b1", c("Mean", "Lower95%CI", "Upper95%CI")])
```

```
##           Mean Lower95%CI Upper95%CI
## 0.2488852 0.2335400 0.2653637
```

18. Proxy for dead — gap analysis

```
# Use a separate object to avoid overwriting data used in BaSTA analysis
# Remove 2023-2024 as too recent to assess the "returned after 2 years" pattern
data_gaps <- data %>%
  filter(year <= 2022) %>%
  # Retain rows with complete data on columns needed for gap analysis only
  filter(!is.na(femid) & !is.na(year) & !is.na(site)) %>%
  arrange(femid, site, year)

# Calculate year gaps for each individual
gap_analysis <- data_gaps %>%
  group_by(femid, site) %>%
  arrange(year) %>%
  mutate(
    next_year = lead(year),
    year_gap = next_year - year,
    is_last_observation = is.na(next_year)) %>%
  ungroup()

# Identify females that missed 2+ years but then returned
# year_gap >= 3: seen in year X and year X+3 means years X+1 and X+2 were missed
returned_after_2yrs <- gap_analysis %>%
  filter(year_gap >= 3) %>%
  select(femid, site, year, next_year, year_gap)

# Summary by site
summary_stats <- gap_analysis %>%
  group_by(site) %>%
  summarise(
    total_females      = n_distinct(femid),
    total_observations = n(),
    mean_gap           = mean(year_gap[!is_last_observation], na.rm = TRUE),
    median_gap         = median(year_gap[!is_last_observation], na.rm = TRUE),
    max_gap            = max(year_gap[!is_last_observation], na.rm = TRUE),
```



```

n_gaps_2plus_missed = sum(year_gap >= 3, na.rm = TRUE), # 2+ years missed
n_gaps_1yr_missed   = sum(year_gap == 2, na.rm = TRUE), # 1 year missed
prop_consecutive    = sum(year_gap == 1, na.rm = TRUE) / sum(!is_last_observation))

print(summary_stats)

```

```

## # A tibble: 2 x 9
##   site total_females total_observations mean_gap median_gap max_gap
##   <fct>      <int>          <int>      <dbl>      <int>    <int>
## 1 CP          1087            2262      1.14          1        5
## 2 OR           972            2199      1.16          1        5
## # i 3 more variables: n_gaps_2plus_missed <int>, n_gaps_1yr_missed <int>,
## #   prop_consecutive <dbl>

```

```

cat("\nNumber of instances where females returned after 2+ years missing:\n")

```

```

##
## Number of instances where females returned after 2+ years missing:

```

```

print(table(returned_after_2yrs$site))

```

```

##
## CP OR
## 27 26

```

```

cat("\nFemales that returned after missing 2+ years:\n")

```

```

##
## Females that returned after missing 2+ years:

```

```

print(returned_after_2yrs)

```

```

## # A tibble: 53 x 5
##   femid site  year next_year year_gap
##   <fct> <fct> <int>    <int>    <int>
## 1 6     OR    2000     2003      3
## 2 193   OR    2001     2006      5
## 3 1156  CP    2001     2004      3
## 4 168   OR    2002     2005      3
## 5 204   OR    2002     2005      3
## 6 231   OR    2002     2005      3
## 7 232   OR    2002     2006      4
## 8 1117  CP    2002     2005      3
## 9 186   OR    2003     2006      3
## 10 338  OR    2003     2007      4
## # i 43 more rows

```

```
# Recapture probability
recapture_prob <- gap_analysis %>%
  filter(!is_last_observation) %>%
  group_by(site) %>%
  summarise(
    prob_recap_next_year = sum(year_gap == 1) / n(),
    prob_recap_within_2yrs = sum(year_gap <= 2) / n())

cat("\nRecapture probabilities:\n")
```

```
##
## Recapture probabilities:
```

```
print(recapture_prob)
```

```
## # A tibble: 2 x 3
##   site prob_recap_next_year prob_recap_within_2yrs
##   <fct>          <dbl>          <dbl>
## 1 CP              0.883              0.977
## 2 OR              0.864              0.979
```

```
# Distribution of maximum gap before returning, per individual
max_gaps <- gap_analysis %>%
  filter(!is_last_observation) %>%
  group_by(femid, site) %>%
  summarise(max_gap_before_return = max(year_gap, na.rm = TRUE)) %>%
  ungroup()

gap_distribution <- max_gaps %>%
  group_by(site) %>%
  count(max_gap_before_return) %>%
  mutate(prop = n / sum(n))

cat("\nDistribution of maximum gaps before returning:\n")
```

```
##
## Distribution of maximum gaps before returning:
```

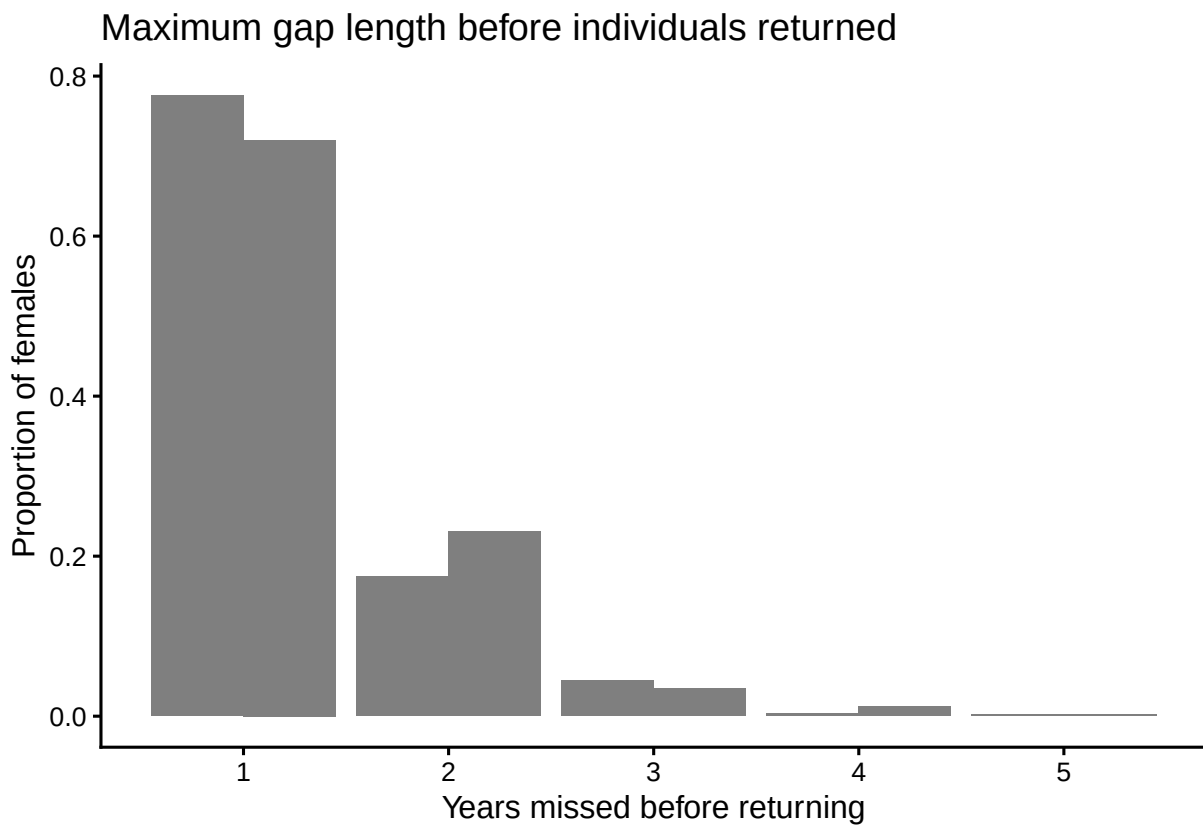
```
print(gap_distribution)
```

```
## # A tibble: 10 x 4
## # Groups:   site [2]
##   site max_gap_before_return     n  prop
##   <fct>          <int> <int>  <dbl>
## 1 CP              1    418 0.776
## 2 CP              2     94 0.174
## 3 CP              3     24 0.0445
## 4 CP              4      2 0.00371
## 5 CP              5      1 0.00186
## 6 OR              1    350 0.720
## 7 OR              2    112 0.230
```

```
## 8 OR      3    17 0.0350
## 9 OR      4     6 0.0123
## 10 OR     5     1 0.00206
```

```
# Visualization
p_gaps <- ggplot(gap_distribution,
                 aes(x = max_gap_before_return, y = prop, fill = site)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = site_colors) +
  labs(x = "Years missed before returning",
       y = "Proportion of females",
       title = "Maximum gap length before individuals returned") +
  theme_publication +
  theme(legend.position = "top")

p_gaps
```



```
ggsave(here("outputs", "figures", "Figure_gap_analysis.png"),
       p_gaps, width = 6, height = 5, dpi = 300)
ggsave(here("outputs", "figures", "Figure_gap_analysis.pdf"),
       p_gaps, width = 6, height = 5)

# Percentages for supplementary text
gap_analysis %>%
  filter(!is_last_observation) %>%
```

```
group_by(site) %>%
summarise(
  total_observable_gaps = n(),
  gaps_2plus_yrs_missed = sum(year_gap >= 3),
  percent_2plus         = round((sum(year_gap >= 3) / n()) * 100, 1))
```

```
## # A tibble: 2 x 4
##   site total_observable_gaps gaps_2plus_yrs_missed percent_2plus
##   <fct>          <int>          <int>          <dbl>
## 1 CP              1175              27            2.3
## 2 OR              1227              26            2.1
```

Temporal pattern of returns

```
returned_after_2yrs %>%
group_by(site) %>%
summarise(
  earliest_year = min(year),
  latest_year   = max(year),
  median_year   = median(year))
```

```
## # A tibble: 2 x 4
##   site earliest_year latest_year median_year
##   <fct>          <int>          <int>          <dbl>
## 1 CP              2001            2019            2013
## 2 OR              2000            2018            2006
```